STACKS, MODULI OF CURVES: LEMMA-GEOMETRICALLY-REDUCED-CONNECTED-1-CURVES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1285-1306 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 1308-1340, the complete author proof (31 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 13 of queue_moduli-curves.json (pb=1262, stmt 1285-1306, proof 1308-1340).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0E1F: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1008–1019. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E0I: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 967–976. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E0G: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1221–1232. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E1G: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1261–1264. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04MH: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1340–1340. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E0E: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 677–688. Cited by the retained statement or proof but not

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part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E0D: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1340–1340. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *There exist an open substack $\mathcal{C}urves^{grc,1} \subset \mathcal{C}urves$ such that*

(1) *given a family of curves $X \to S$ the following are equivalent*
   (a) *the classifying morphism $S \to \mathcal{C}urves$ factors through $\mathcal{C}urves^{grc,1}$,*
   (b) *the geometric fibres of the morphism $X \to S$ are reduced, connected, and have dimension 1,*

(2) *given a scheme X proper over a field k with $\dim(X) \leq 1$ the following are equivalent*
   (a) *the classifying morphism $\mathrm{Spec}(k) \to \mathcal{C}urves$ factors through $\mathcal{C}urves^{grc,1}$,*
   (b) *X is geometrically reduced, geometrically connected, and has dimension 1.*

Proof. By Lemmas 0E0G, 0E1G, 0E0I, and 0E1F it is clear that we have

$$\mathcal{C}urves^{grc,1} \subset \mathcal{C}urves^{geomred} \cap \mathcal{C}urves^{CM,1}$$

if it exists. Let $f : X \to S$ be a family of curves such that f is Cohen-Macaulay, has geometrically reduced fibres, and has relative dimension 1. By More on Morphisms of Spaces, Lemma 0E0D in the Stein factorization

$$X \to T \to S$$

the morphism $T \to S$ is étale. This implies that there is an open and closed subscheme $S' \subset S$ such that $X \times_S S' \to S'$ has geometrically connected fibres (in the decomposition of Morphisms, Lemma 04MH for the finite locally free morphism $T \to S$ this corresponds to S_1). Formation of this open commutes with arbitrary base change because the number of connected components of geometric fibres is invariant under base change (it is also true that the Stein factorization commutes with base change in our particular case but we don't need this to conclude). Thus we get the open substack with the desired properties by the method discussed in Section 0E0E. □